Meeting 4A — Runtime State Architecture

Part 1: System Runtime States & UI State Machine

Purpose

This document defines the **actual software runtime states** of the Digital Harmonium Workstation.

These are **true software states** that will exist inside the firmware.

Only one state from each state machine may be active at a time.

Every state has:

- Purpose
- Entry Conditions
- Responsibilities
- Valid Transitions
- Exit Conditions

These states will eventually become C++ enums and state managers inside the ESP32 firmware.

Runtime State Machine 1 — System Runtime

This is the highest level state machine.

It controls the entire workstation.

S1 — POWER_OFF

Purpose

No firmware execution.

Entire system is unpowered.

Entry

Power removed.

Responsibilities

None.

Valid Transition

→ BOOTING

Exit

Power switch turned ON.

S2 — BOOTING

Purpose

Initialize ESP32 hardware.

Entry

Power applied.

Responsibilities

- Initialize CPU
- Initialize RAM
- Initialize Flash
- Configure GPIO
- Initialize timers
- Configure interrupts

Valid Transition

→ SELF_TEST

Exit

All hardware initialized.

S3 — SELF_TEST

Purpose

Verify hardware before operation.

Entry

BOOTING completed.

Responsibilities

- Test LEDs
- Test display
- Test audio codec
- Verify SD card
- Verify keyboard matrix
- Verify amplifier

Valid Transition

→ LOADING_SYSTEM

Exit

Hardware tests completed.

S4 — LOADING_SYSTEM

Purpose

Load persistent system resources.

Entry

Self-test passed.

Responsibilities

- Load configuration
- Mount SD card
- Load default settings
- Prepare runtime managers

Valid Transition

→ IDLE

Exit

System ready.

S5 — IDLE

Purpose

Normal waiting state.

Entry

System initialized.

Responsibilities

- Scan keyboard
- Scan buttons
- Refresh LCD
- Wait for events

Valid Transitions

→ MENU_ACTIVE

→ PLAYING

→ RECORDING

→ PLAYBACK

→ SHUTDOWN

Exit

User requests another mode.

S6 — SHUTDOWN

Purpose

Safely terminate operation.

Entry

Shutdown requested.

Responsibilities

- Stop audio
- Flush recordings
- Save settings
- Unmount SD

Valid Transition

→ POWER_OFF

Exit

Power removed.

Runtime State Machine 2 — User Interface

The UI always operates independently from audio generation.

Changing menus must never interrupt sound.

U1 — HOME_SCREEN

Purpose

Primary operating screen.

Entry

System reaches IDLE.

Responsibilities

Display:

- Current Instrument
- Active transpose
- Status indicators
- Small runtime icons

Valid Transitions

→ INSTRUMENT_MENU

→ TABLA_MENU

→ TANPURA_MENU

→ METRONOME_MENU

→ RECORD_MENU

→ SETTINGS_MENU

Exit

User selects another menu.

U2 — INSTRUMENT_MENU

Purpose

Instrument selection.

Entry

Instrument button pressed.

Responsibilities

- Browse instruments
- Display instrument name
- Preview selection

Valid Transition

→ INSTRUMENT_PARAMETERS

→ HOME_SCREEN

Exit

Instrument selected.

U3 — INSTRUMENT_PARAMETERS

Purpose

Adjust currently selected instrument.

Entry

Instrument selected.

Responsibilities

- Change transpose
- Display root key
- Show active effects
- Display instrument information

Valid Transition

→ HOME_SCREEN

Exit

User returns.

U4 — TABLA_MENU

Purpose

Control tabla accompaniment.

Entry

Tabla button pressed.

Responsibilities

- Select taal
- Tempo adjustment
- Start/Stop tabla

Valid Transition

→ HOME_SCREEN

Exit

Return requested.

U5 — TANPURA_MENU

Purpose

Control tanpura drone.

Entry

Tanpura button pressed.

Responsibilities

- Start/Stop
- Drone pitch
- Volume
- Tempo (if applicable)

Valid Transition

→ HOME_SCREEN

Exit

Return requested.

U6 — METRONOME_MENU

Purpose

Practice timing control.

Entry

Metronome selected.

Responsibilities

- Tempo adjustment
- Enable/Disable
- Beat selection

Valid Transition

→ HOME_SCREEN

Exit

Return requested.

U7 — RECORD_MENU

Purpose

Recording control interface.

Entry

Record button pressed.

Responsibilities

Display:

- Recording status
- Elapsed time
- File name
- Storage availability

Valid Transition

→ RECORDING_SCREEN

→ HOME_SCREEN

Exit

Recording started or cancelled.

U8 — RECORDING_SCREEN

Purpose

Live recording display.

Entry

Recording begins.

Responsibilities

- Timer
- Status indicator
- File size
- Recording icon

Valid Transition

→ PLAYBACK_MENU

→ HOME_SCREEN

Exit

Recording stops.

U9 — PLAYBACK_MENU

Purpose

Playback management.

Entry

Playback button pressed.

Responsibilities

- File selection
- Play
- Pause
- Delete
- Rename (future)

Valid Transition

→ PLAYBACK_SCREEN

→ HOME_SCREEN

Exit

Playback begins.

U10 — PLAYBACK_SCREEN

Purpose

Display playback progress.

Entry

Playback starts.

Responsibilities

- Playback timer
- Current file
- Progress indicator
- LED keyboard guidance

Valid Transition

→ HOME_SCREEN

Exit

Playback completed or stopped.

U11 — SETTINGS_MENU

Purpose

Global workstation configuration.

Entry

Settings selected.

Responsibilities

- LCD contrast
- Speaker volume
- System information

- Factory reset

Valid Transition

→ SYSTEM_INFO

→ HOME_SCREEN

Exit

Settings saved.

U12 — SYSTEM_INFO

Purpose

Display firmware information.

Entry

System Info selected.

Responsibilities

Show:

- Firmware version
- Build version
- RAM usage
- CPU usage
- SD status
- Instrument count

Valid Transition

→ SETTINGS_MENU

Exit

Back requested.

Summary

This document defines **17 actual runtime states** that will be explicitly implemented inside the firmware:

System Runtime

- POWER_OFF
- BOOTING
- SELF_TEST
- LOADING_SYSTEM
- IDLE
- SHUTDOWN

User Interface

- HOME_SCREEN
- INSTRUMENT_MENU
- INSTRUMENT_PARAMETERS
- TABLA_MENU
- TANPURA_MENU
- METRONOME_MENU
- RECORD_MENU
- RECORDING_SCREEN
- PLAYBACK_MENU
- PLAYBACK_SCREEN
- SETTINGS_MENU
- SYSTEM_INFO

These are **real software states** and will later become dedicated state managers inside the ESP32 runtime.